IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Project repository: https://github.com/MateoGomezTamayo/Data-Science-knowledge-base/tree/main/10_portfolio/21_spacex_capstone

Executive Summary

- **Goal:** predict whether a Falcon 9 first stage lands, to estimate launch cost
- **Data:** SpaceX API, web scraping and SQL records
- **Methods:** wrangling, EDA, Folium maps, a Plotly Dash app and four tuned classifiers
- **Result:** landing success climbed from 0% in 2013 to 84% in 2020
- **Model:** all four tie on the test set, so the choice needs a stronger test

66.7%

of 90 Falcon 9 flights landed successfully

83.3%

test accuracy for all four models (15 of 18)

66.7%

baseline: always predict a landing on the test set

5.6 pts

accuracy change from one test error

A test set of 18 flights cannot separate these models, so I reran them on 100 random splits.

Introduction

- SpaceX advertises Falcon 9 launches at $62M, while other providers charge $165M or more
- Much of that saving comes from reusing the first stage
- If we can predict a landing, we can estimate the cost of a launch and bid against SpaceX

Questions this project answers

1. Which factors are linked to a successful landing?
2. How do launch sites compare, and where are they located?
3. Can a model predict the outcome reliably?
4. How much can we trust that prediction on 90 flights?

Section 1

Methodology



Methodology

Data collection: SpaceX REST API data and Wikipedia launch tables

Data wrangling: Falcon 9 filter, payload imputation, binary Class target, one-hot encoding

EDA: visualization with Seaborn and Matplotlib, and SQL on SQLite

Interactive analytics: Folium maps and a Plotly Dash dashboard

Predictive analysis: grid search with 10-fold CV on four classifiers, then 100 random splits

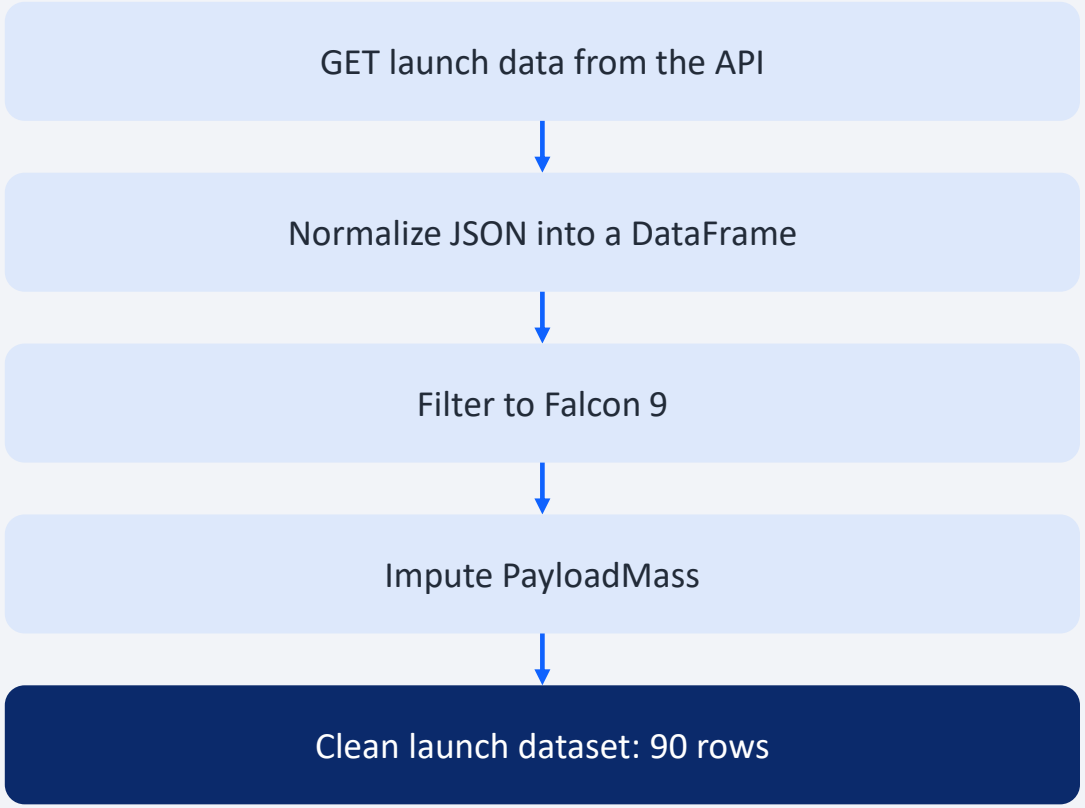
Data Collection

- **SpaceX API:** rocket, payload, launchpad and core details as JSON
- **Wikipedia:** Falcon 9 launch tables parsed from a fixed page revision
- **IBM files:** the prepared SQL, geolocation and dashboard datasets used in the labs
- **Note:** the live API was down during this work, so the notebook uses IBM's static copy of the response

GitHub: https://github.com/MateoGomezTamayo/Data-Science-knowledge-base/tree/main/10_portfolio/21_spacex_capstone

Data Collection – SpaceX API

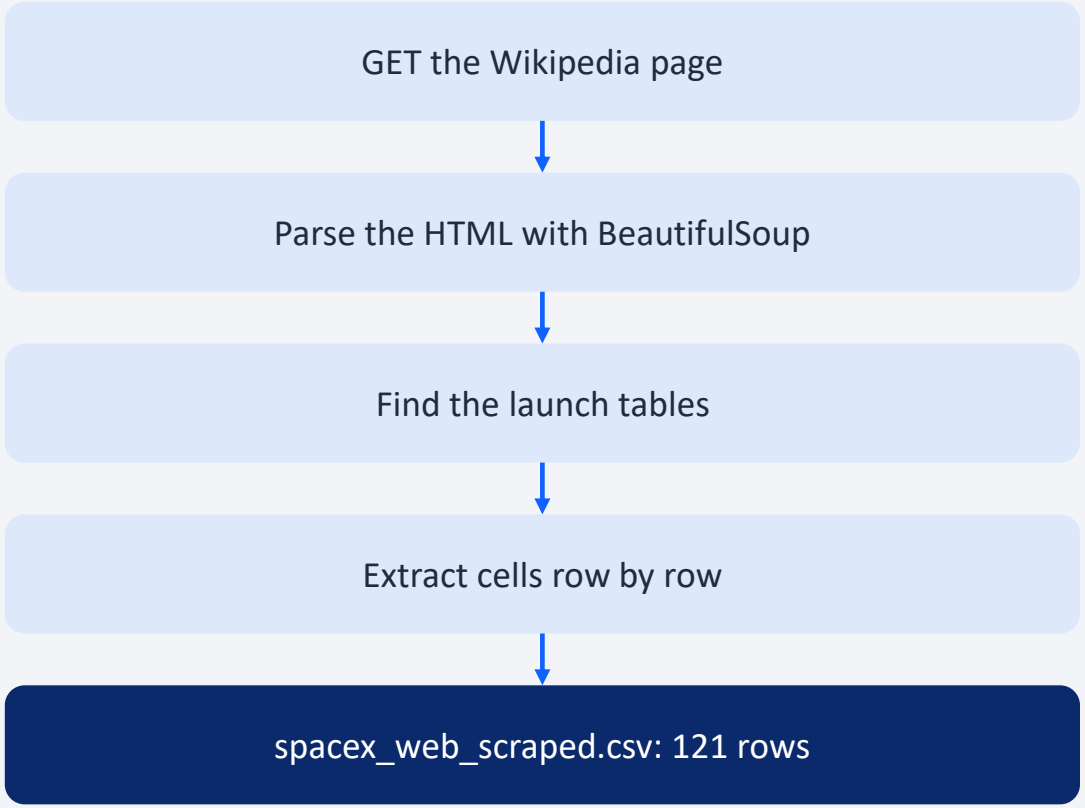
- GET request for past launches
- Parse the JSON with json_normalize
- Keep Falcon 9 launches only: 90 of 107
- Fill missing payloads with the mean: 6,104.96 kg



GitHub: https://github.com/MateoGomezTamayo/Data-Science-knowledge-base/blob/main/10_portfolio/19_spacex_data_collection_wrangling/analysis.ipynb

Data Collection – Scraping

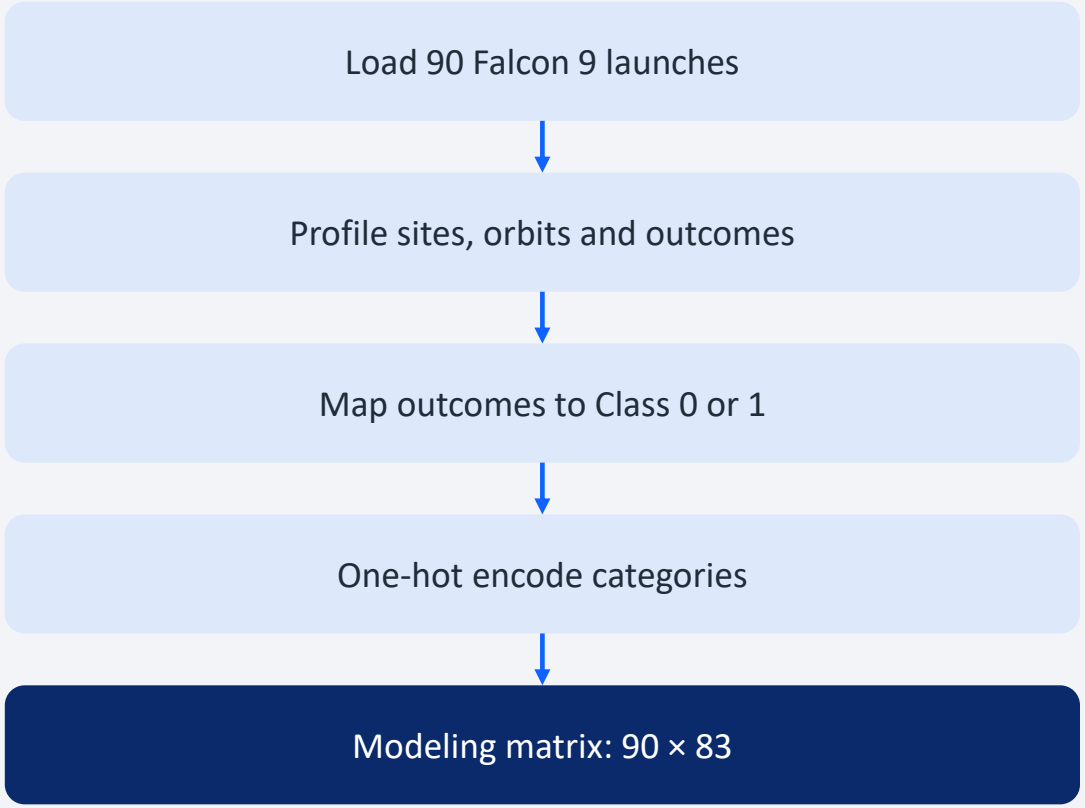
- Request a fixed Wikipedia revision, so results can be repeated
- Find the launch tables with BeautifulSoup: 9 tables
- Extract 11 columns from each launch row
- Result: 121 launch records saved as CSV



GitHub: https://github.com/MateoGomezTamayo/Data-Science-knowledge-base/blob/main/10_portfolio/21_spacex_capstone/01_web_scraping.ipynb

Data Wrangling

- Count launches by site, orbit and outcome
- Mark 'True ASDS', 'True RTLS' and 'True Ocean' as successful landings
- Create Class: 1 = landed, 0 = did not
- Overall rate: 60 of 90 flights, 66.7%
- One-hot encode categories into 83 features for modeling



GitHub: https://github.com/MateoGomezTamayo/Data-Science-knowledge-base/blob/main/10_portfolio/19_spacex_data_collection_wrangling/analysis.ipynb

EDA with Data Visualization

- Each chart answers one question about landing success.

Chart	Why it was used
Scatter: flight number vs. site	Shows whether sites improve with experience
Scatter: payload vs. site	Shows which payloads each site handles
Bar: success rate by orbit	Compares orbit types, with launch counts
Scatter: flight number vs. orbit	Checks for a learning effect in each orbit
Scatter: payload vs. orbit	Checks how payload relates to each orbit
Line: yearly success rate	Shows the overall trend over time

GitHub: https://github.com/MateoGomezTamayo/Data-Science-knowledge-base/blob/main/10_portfolio/21_spacex_capstone/03_eda_visualization.ipynb

EDA with SQL

- Listed the distinct launch sites
- Filtered records for sites beginning with CCA
- Summed payload for NASA (CRS) and averaged it for F9 v1.1
- Found the first successful ground-pad landing
- Listed drone-ship successes with 4,000–6,000 kg payloads
- Counted mission outcomes and found the maximum-payload boosters
- Listed 2015 drone-ship failures and ranked landing outcomes

GitHub: https://github.com/MateoGomezTamayo/Data-Science-knowledge-base/blob/main/10_portfolio/21_spacex_capstone/02_eda_sql.ipynb

Build an Interactive Map with Folium

- **Circles and labels:** mark all four launch sites
- **Marker clusters:** show each launch as green (success) or red (failure)
- **Lines and distance labels:** measure how close a site is to coast, rail, road and city
- **Why:** geography can't be seen in a table, and overlapping launches need clustering

GitHub: https://github.com/MateoGomezTamayo/Data-Science-knowledge-base/blob/main/10_portfolio/21_spacex_capstone/04_interactive_map.ipynb

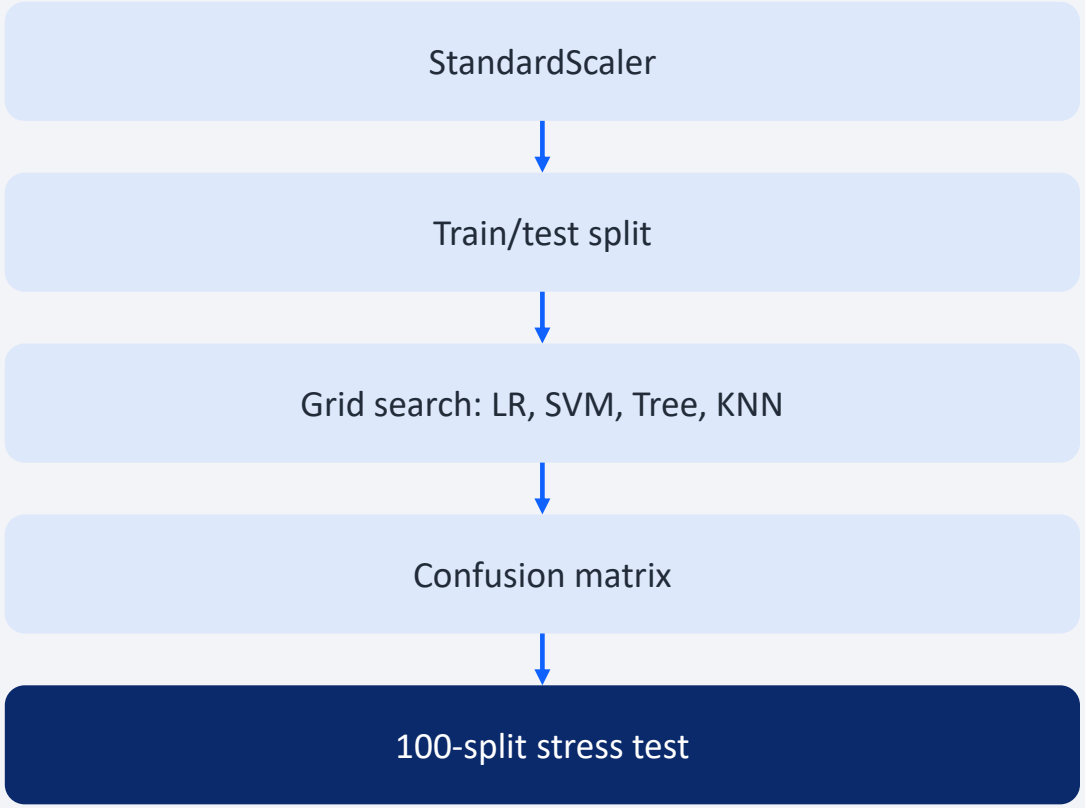
Build a Dashboard with Plotly Dash

- **Dropdown:** choose all sites or one site
- **Pie chart:** share of successes by site, or success vs. failure for one site
- **Range slider:** filter by payload mass
- **Scatter plot:** payload vs. outcome, colored by booster version
- **Why:** lets a reader compare sites and payload ranges without new code

GitHub: https://github.com/MateoGomezTamayo/Data-Science-knowledge-base/blob/main/10_portfolio/21_spacex_capstone/spacex_dash_app.py

Predictive Analysis (Classification)

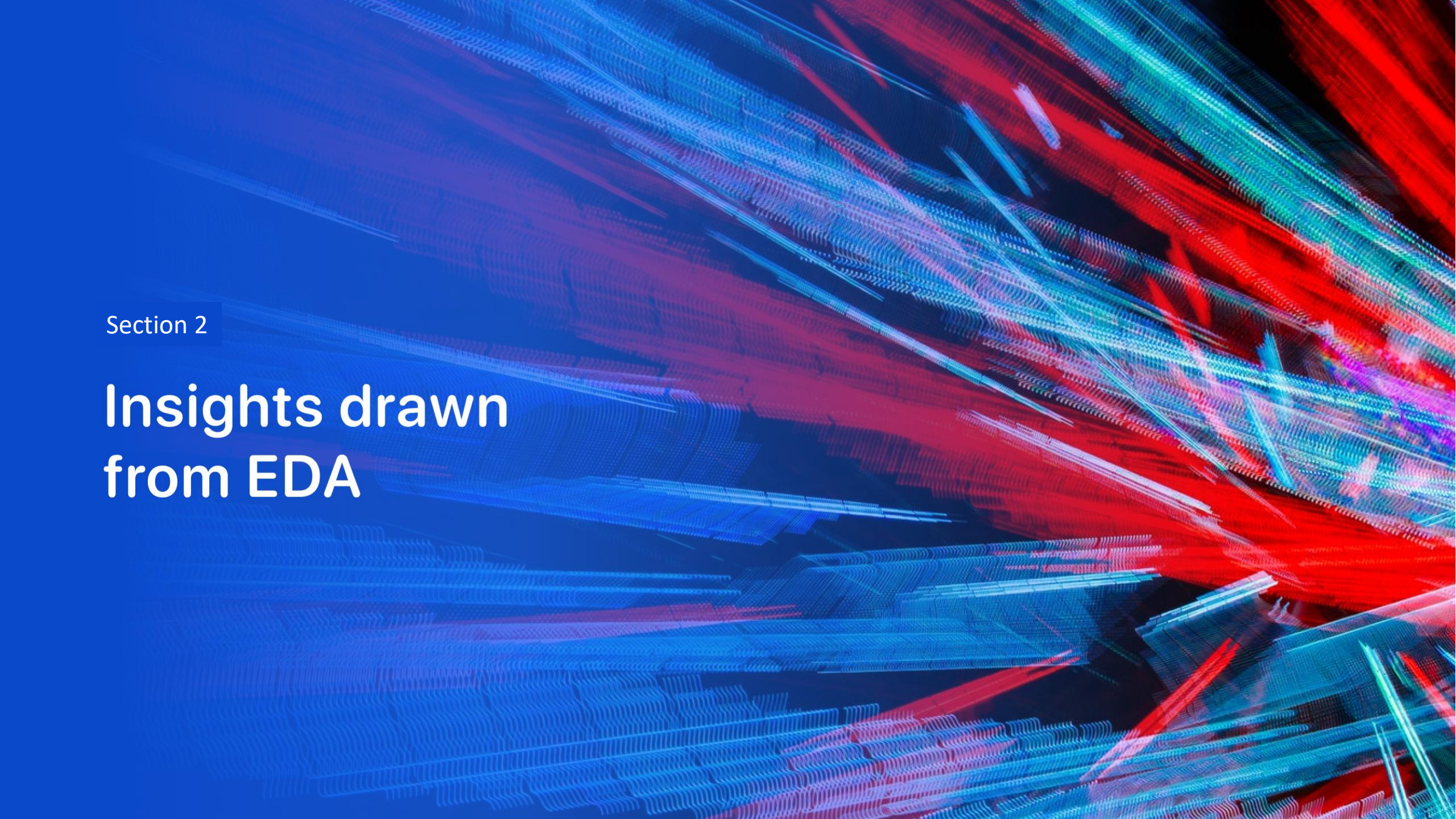
- Standardize the 83 features
- Split 80/20: 72 train, 18 test (random_state=2)
- Tune four models with GridSearchCV, cv=10
- Compare CV and test accuracy
- Stress test on 100 stratified splits



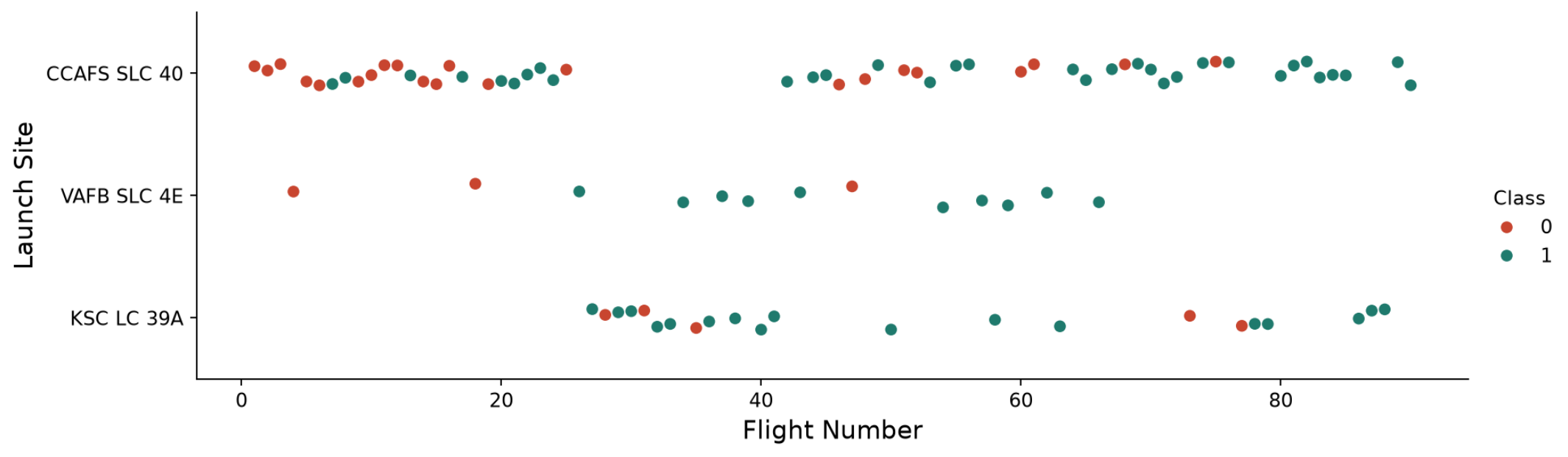
GitHub: https://github.com/MateoGomezTamayo/Data-Science-knowledge-base/blob/main/10_portfolio/21_spacex_capstone/05_classification.ipynb

Results

- **EDA:** success rises with flight number, and most heavy payloads land
- **SQL:** 100 of 101 missions succeeded, but landings are a separate outcome
- **Maps:** all sites sit near the coast, and CCAFS SLC-40 is about 20 km from the city of Cape Canaveral
- **Dashboard:** KSC LC-39A leads in both share of successes and success rate
- **Models:** all four reach 83.3% on the test set

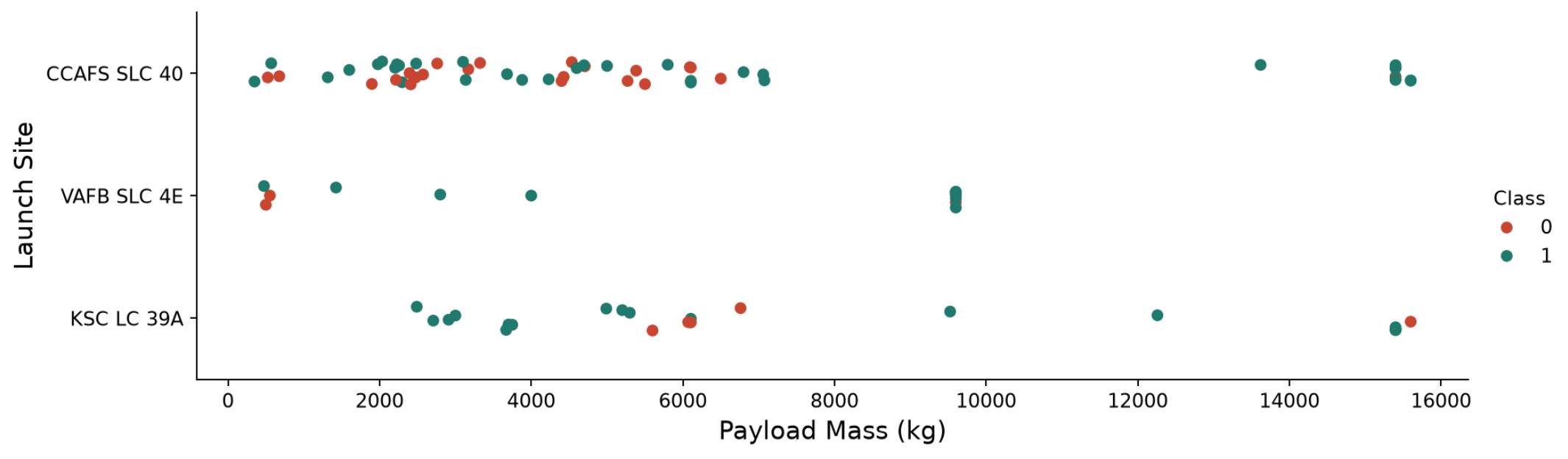

Section 2
Insights drawn from EDA

Flight Number vs. Launch Site



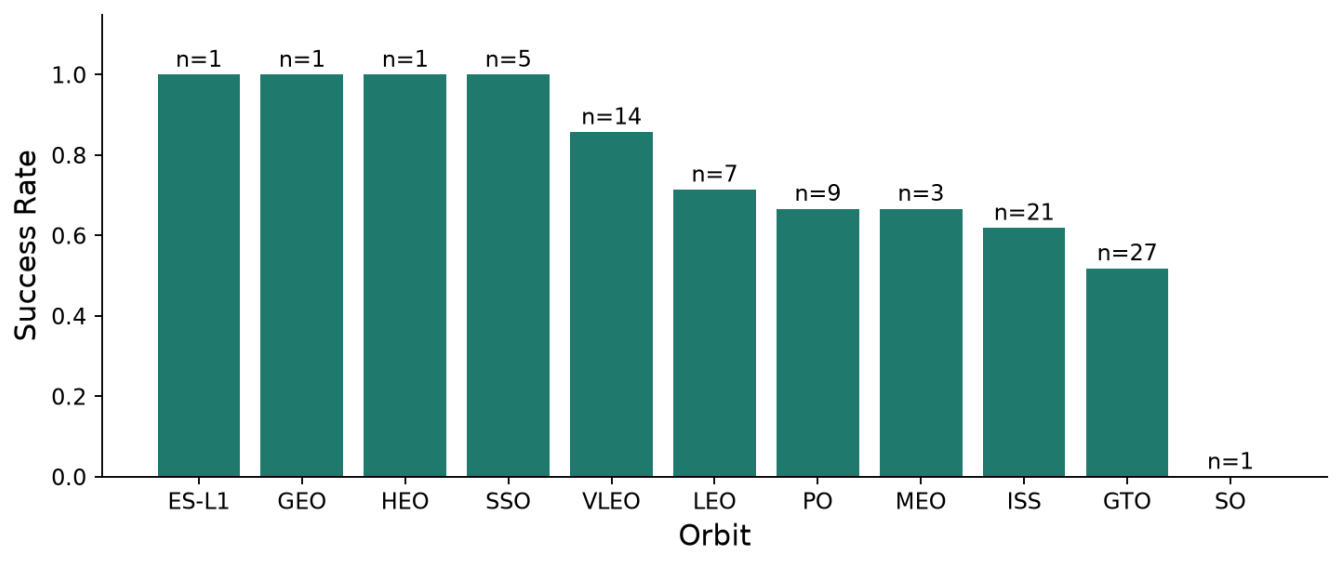
- Success improves with flight number at all three sites (green = landed)
- CCAFS SLC 40 has the lowest rate, 60.0% of 55, partly because it hosted most early flights
- KSC LC 39A reached 77.3% of 22, and VAFB SLC 4E 76.9% of 13

Payload vs. Launch Site



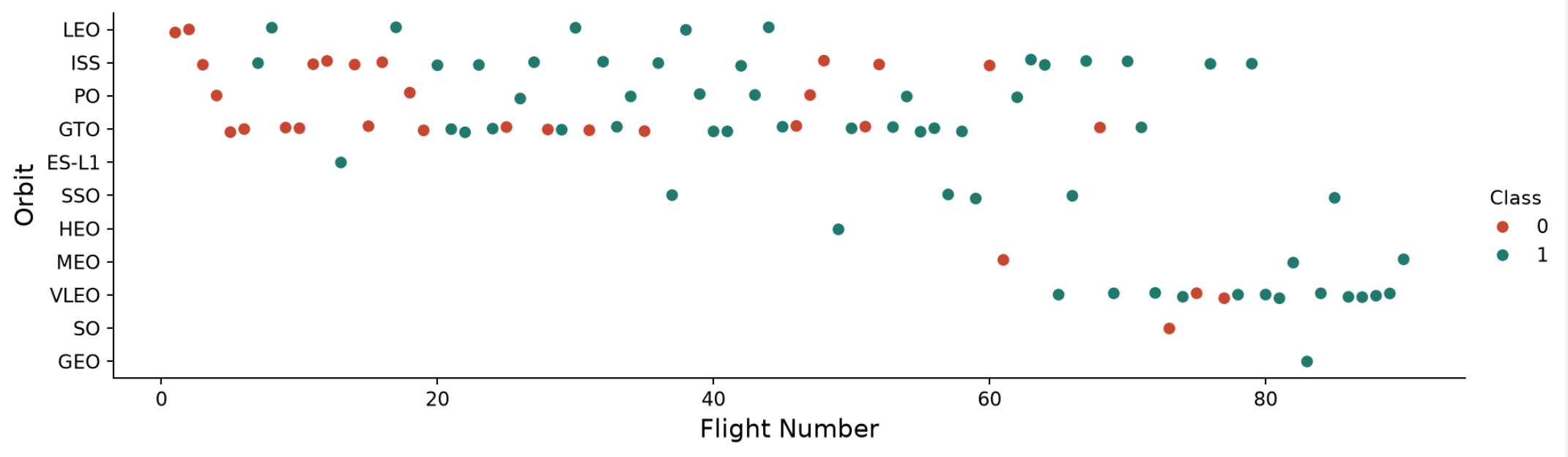
- Heavier flights land more often: above 8,000 kg, 87% of 23; at or below it, 60% of 67
- VAFB SLC 4E has no flights above 10,000 kg in this data
- CCAFS SLC 40 and KSC LC 39A carry the heaviest payloads, near 15,600 kg

Success Rate vs. Orbit Type



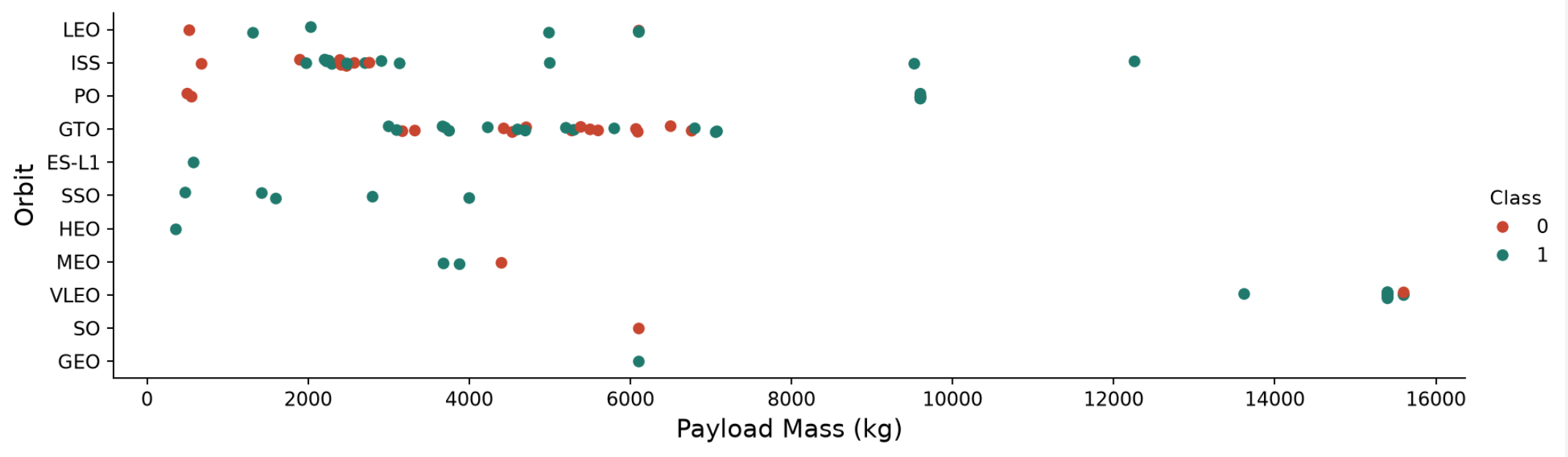
- ES-L1, GEO, HEO and SO have one flight each, so their rates mean little
- SSO: 5 of 5, and VLEO: 85.7% of 14
- GTO: 51.9% of 27, the lowest rate with a solid sample

Flight Number vs. Orbit Type



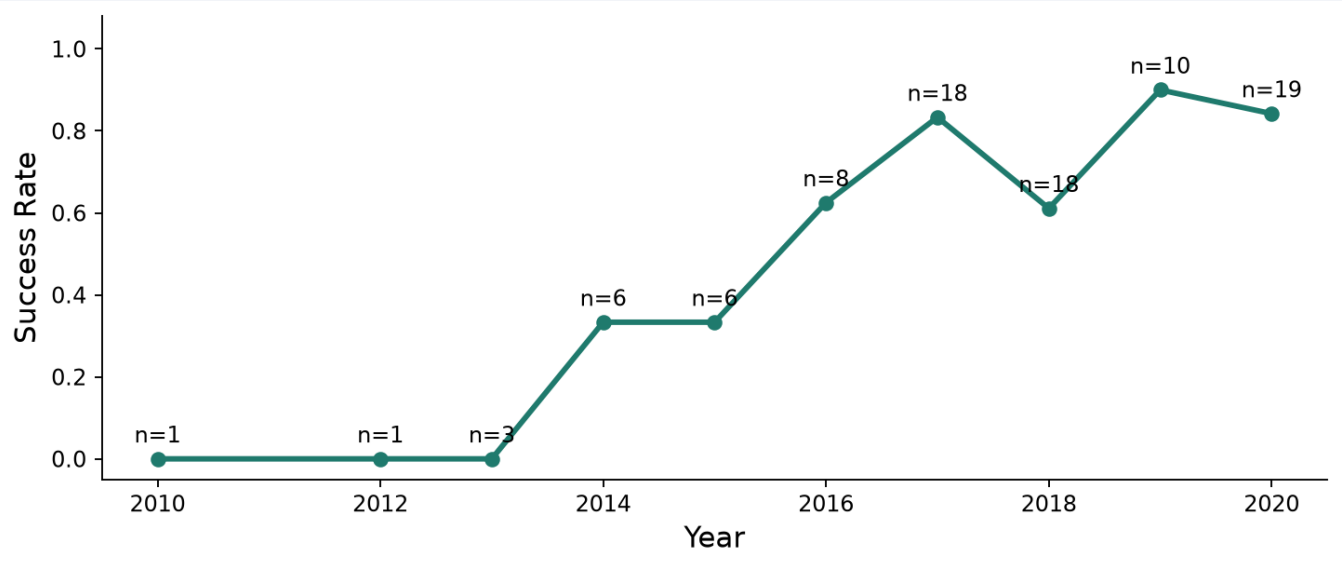
- LEO and ISS failures sit mainly in early flights, and later flights mostly land
- GTO stays mixed: no clear gain with flight number
- VLEO starts at flight 65, so it likely benefits from a more mature program

Payload vs. Orbit Type



- In ISS, every failure is below 2,800 kg, and the heavier flights landed; PO has a failure near 9,600 kg
- GTO mixes successes and failures across its 3,000–7,000 kg range
- VLEO carries the heaviest payloads and mostly lands

Launch Success Yearly Trend



- 0% through 2013
- 33% in 2014 and 2015
- 83% in 2017, then a drop to 61% in 2018
- 90% in 2019 and 84% in 2020
- Early points rest on few flights, so read the n labels

All Launch Site Names

- Four launch sites. CCAFS appears under two names, LC-40 and SLC-40.

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- The earliest CCA records are demo and cargo flights with no landing success.

```
SELECT * FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Booster	Site	Customer	Landing
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	NASA (COTS) NRO	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	NASA (CRS)	No attempt

Total Payload Mass

- NASA's CRS cargo missions carried 45,596 kg in total.

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass
FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

Total_Payload_Mass
45,596 kg

Average Payload Mass by F9 v1.1

- F9 v1.1 boosters carried about 2,535 kg on average.

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass
FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%';
```

Avg_Payload_Mass
2,534.67 kg

First Successful Ground Landing Date

- The first successful ground-pad landing was on December 22, 2015.

```
SELECT MIN(Date) AS First_Ground_Landing FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

First_Ground_Landing
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- Four FT boosters landed on a drone ship with payloads between 4,000 and 6,000 kg.

```
SELECT Booster_Version FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- 100 of 101 missions succeeded. Mission success doesn't mean the booster landed.

```
SELECT CASE WHEN Mission_Outcome LIKE 'Success%'
  THEN 'Success' ELSE 'Failure' END AS Outcome, COUNT(*) AS Total
FROM SPACEXTABLE GROUP BY Outcome;
```

Outcome	Total
Success	100
Failure	1

Boosters Carried Maximum Payload

- Twelve F9 B5 boosters carried the maximum payload, 15,600 kg.

```
SELECT Booster_Version FROM SPACEXTABLE WHERE PAYLOAD_MASS__KG_ =
  (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

F9 B5	F9 B5	F9 B5
B1048.4	B1049.4	B1051.3
B1056.4	B1048.5	B1051.4
B1049.5	B1060.2	B1058.3
B1051.6	B1060.3	B1049.7

2015 Launch Records

- Both 2015 drone-ship failures used F9 v1.1 boosters from CCAFS LC-40.

```
SELECT substr(Date, 6, 2) AS Month, Landing_Outcome,
  Booster_Version, Launch_Site FROM SPACEXTABLE
WHERE Landing_Outcome = 'Failure (drone ship)'
  AND substr(Date, 1, 4) = '2015';
```

Month	Landing outcome	Booster	Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- From 2010 to March 2017, 'No attempt' was the most common outcome, and drone ships had 5 successes and 5 failures.

```
SELECT Landing_Outcome, COUNT(*) AS Total,
 RANK() OVER (ORDER BY COUNT(*) DESC)
  AS Rank
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04'
  AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Total DESC;
```

Landing outcome	Total	Rank
No attempt	10	1
Success (drone ship)	5	2
Failure (drone ship)	5	2
Success (ground pad)	3	4
Controlled (ocean)	3	4
Uncontrolled (ocean)	2	6
Failure (parachute)	2	6
Precluded (drone ship)	1	8

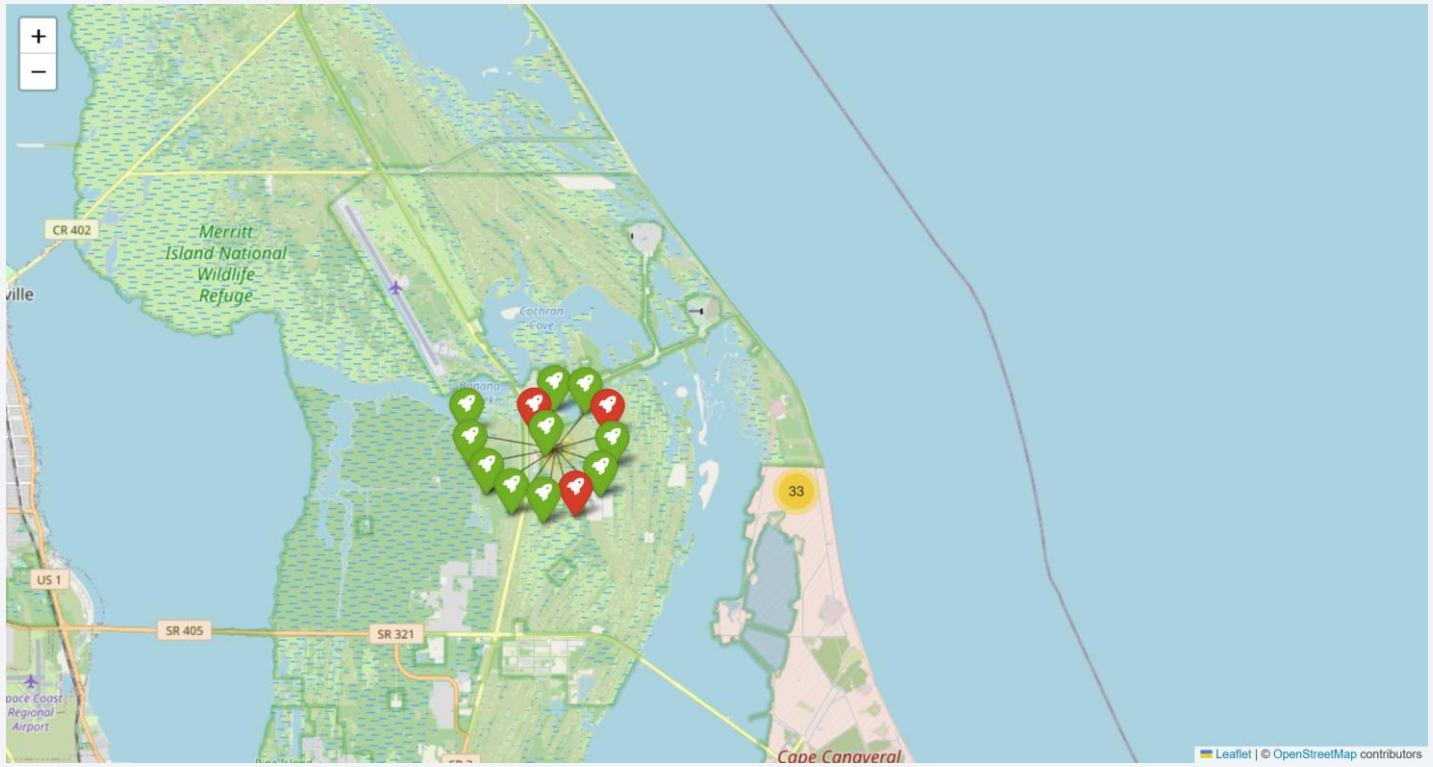

Section 3
Launch Sites
Proximities Analysis

All Launch Sites Are on the Coast



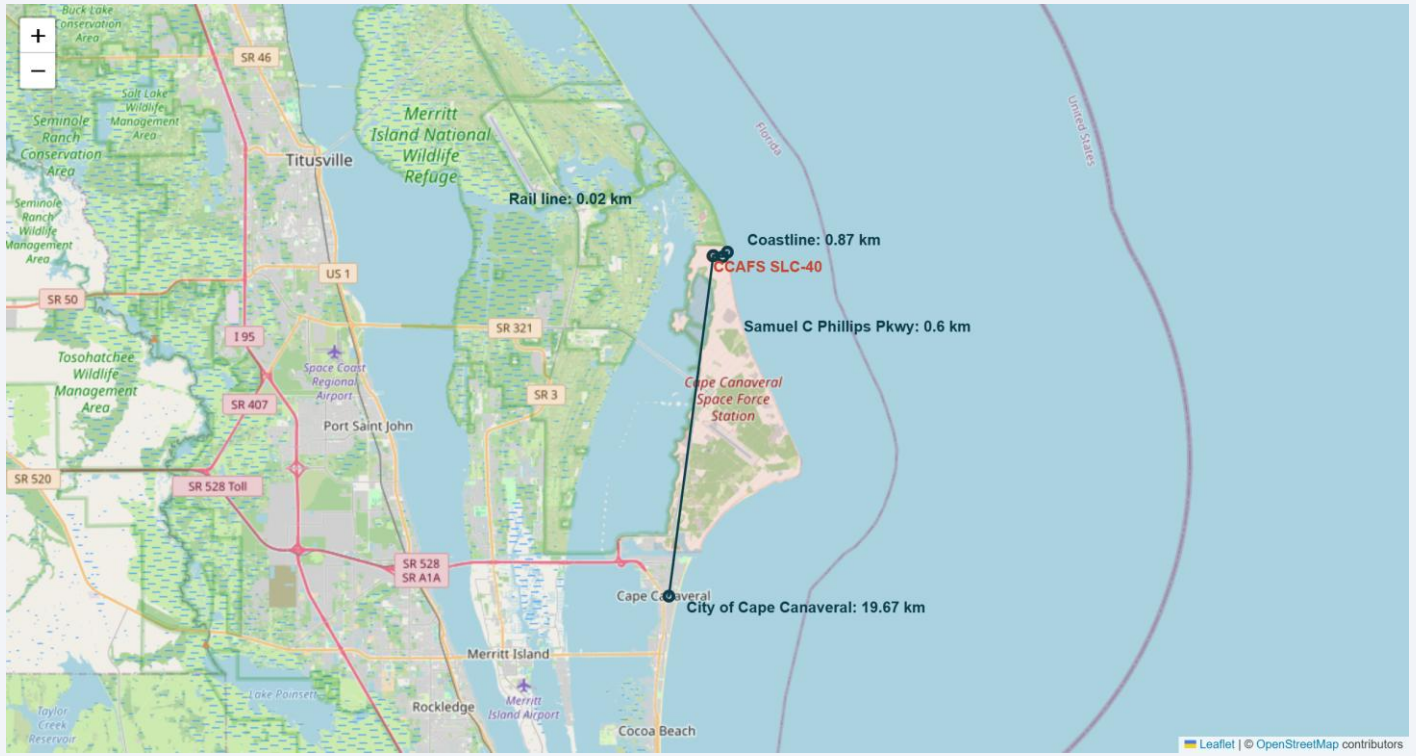
- Three sites are in Florida and one is in California
- All sit near the ocean, in the southern United States
- Launches over water keep failures away from people
- The Florida sites overlap at this zoom level

Launch Outcomes at KSC LC-39A



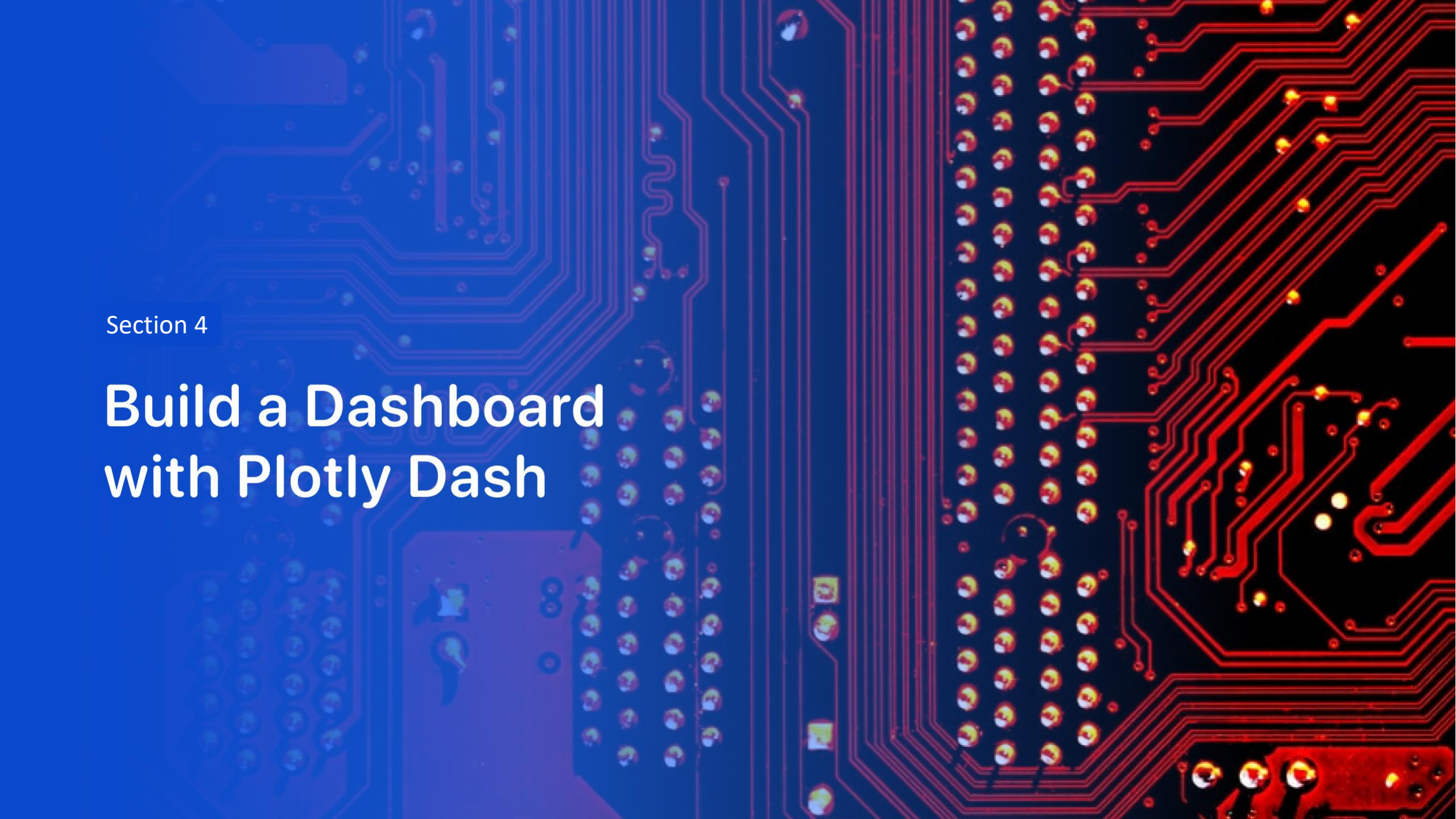
- The opened cluster shows all 13 KSC launches
- Green: 10 landings
- Red: 3 failures
- 76.9% is the best site rate in this dataset
- Launches at one site share coordinates, so clusters are needed

CCAFS SLC-40 Proximities

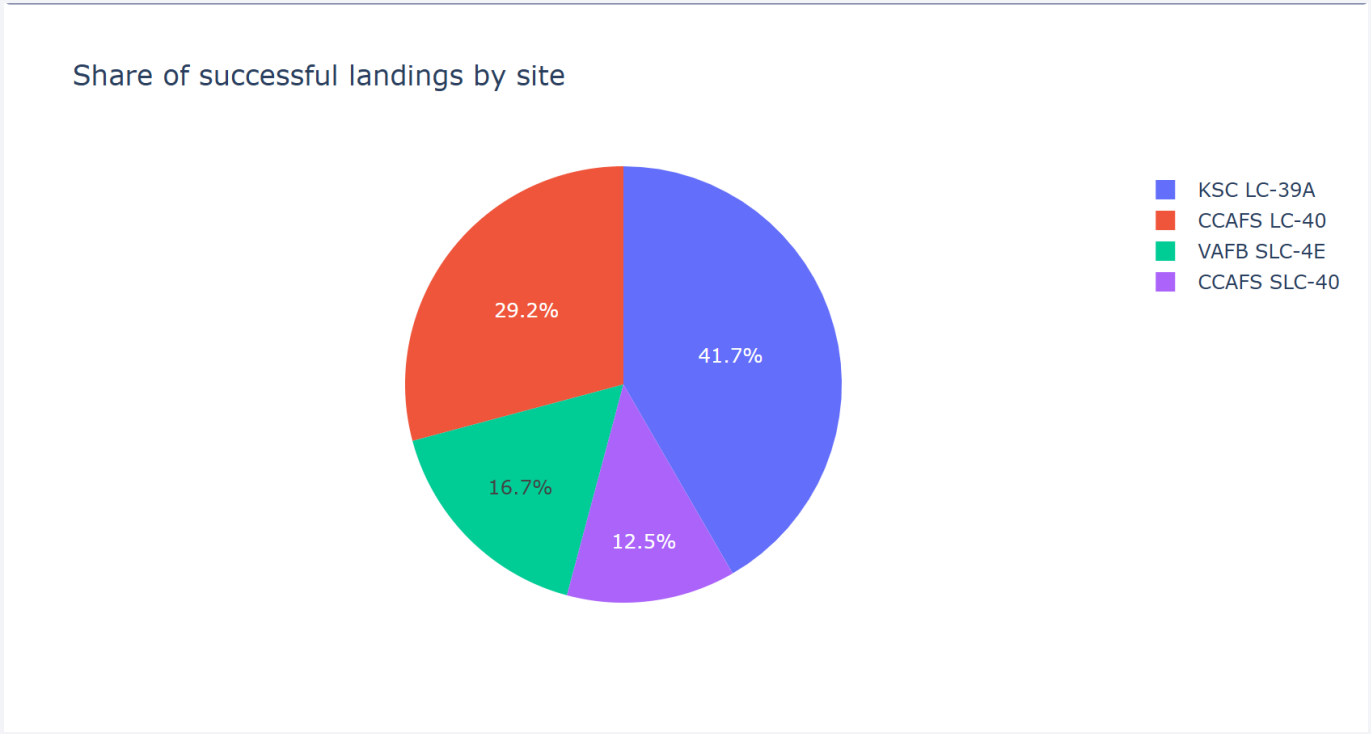


Feature	Distance
Rail line	0.02 km
Samuel C Phillips Pkwy	0.60 km
Coastline	0.87 km
City of Cape Canaveral	19.67 km

The site has rail and road access and sits by the coast, and it is about 20 km from the city of Cape Canaveral.
Distances use OpenStreetMap coordinates.

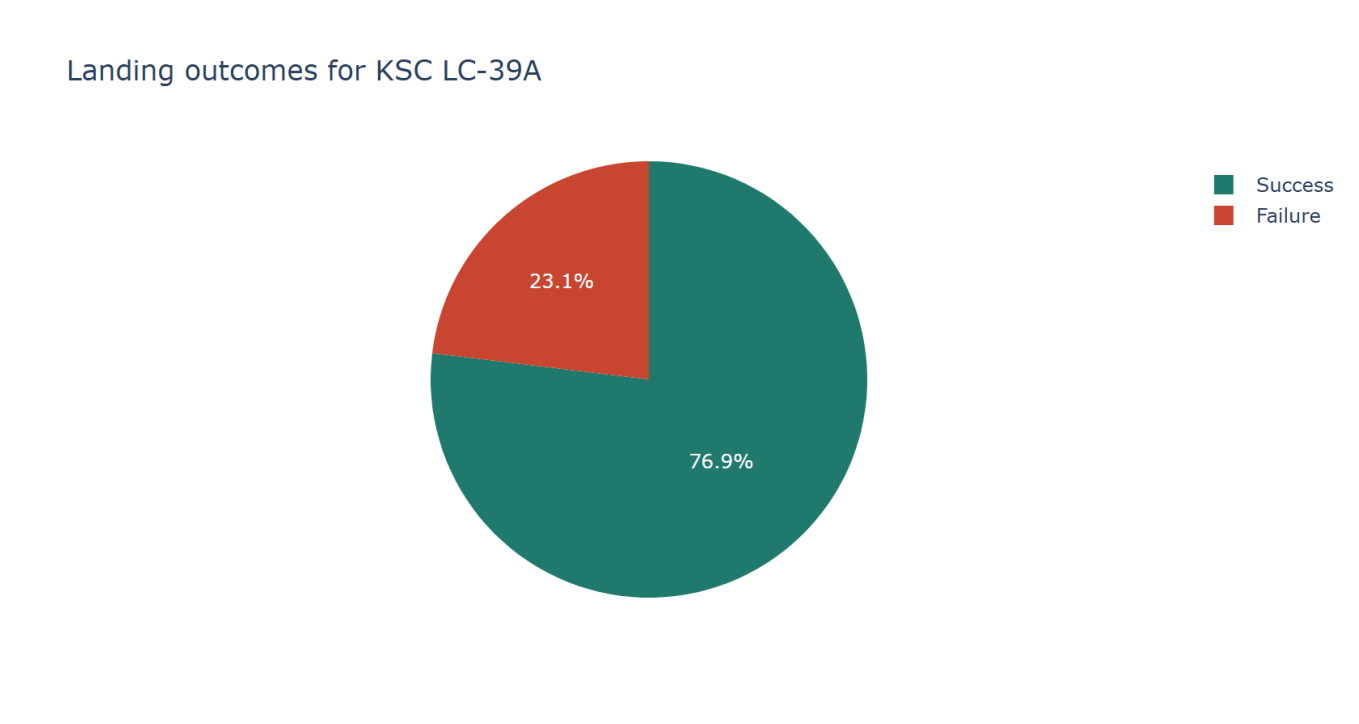

Section 4
Build a Dashboard with Plotly Dash

KSC LC-39A Has the Most Successes



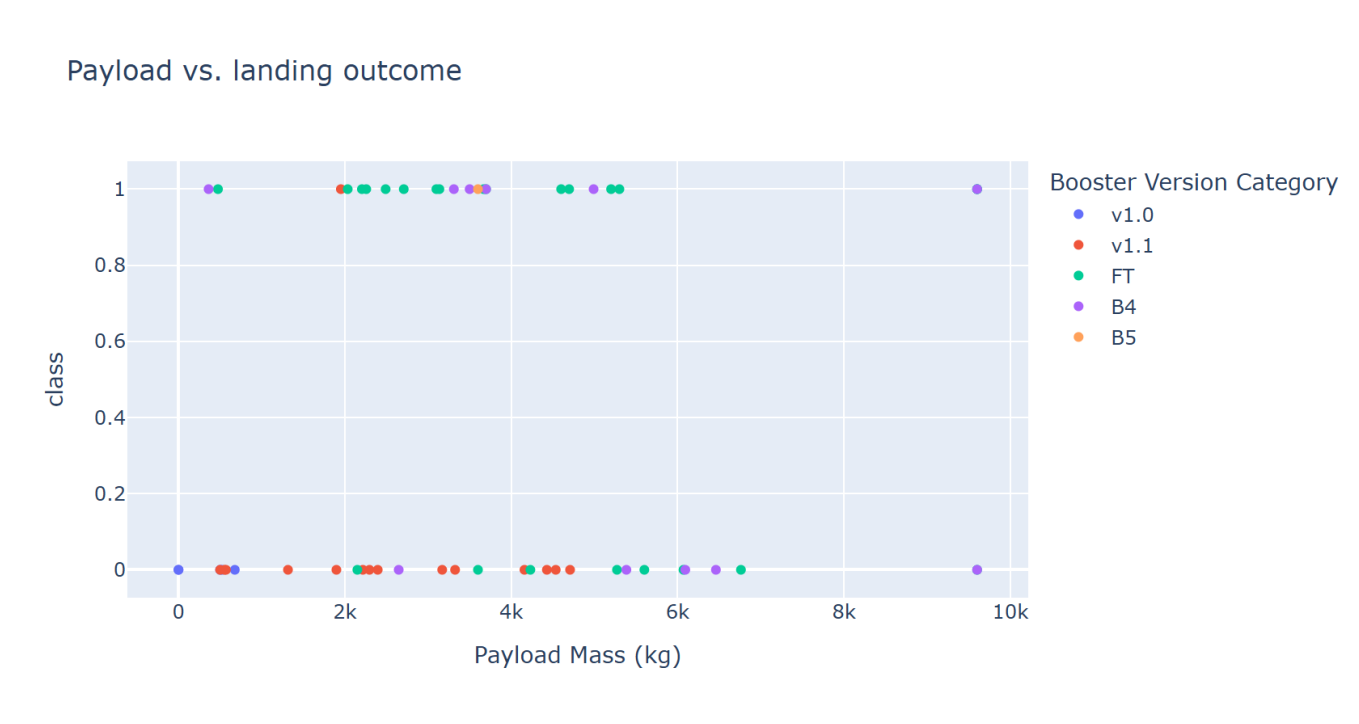
- KSC holds 41.7% of all successes
- CCAFS LC-40 follows with 29.2%
- The dashboard data has 56 flights, a smaller set than the 90 used elsewhere

KSC LC-39A Has the Highest Success Rate



- 10 of 13 launches landed: 76.9%
- The next best site is CCAFS SLC-40 at 42.9% of 7
- With 13 flights, this rate is still uncertain

Payload vs. Landing Outcome by Booster



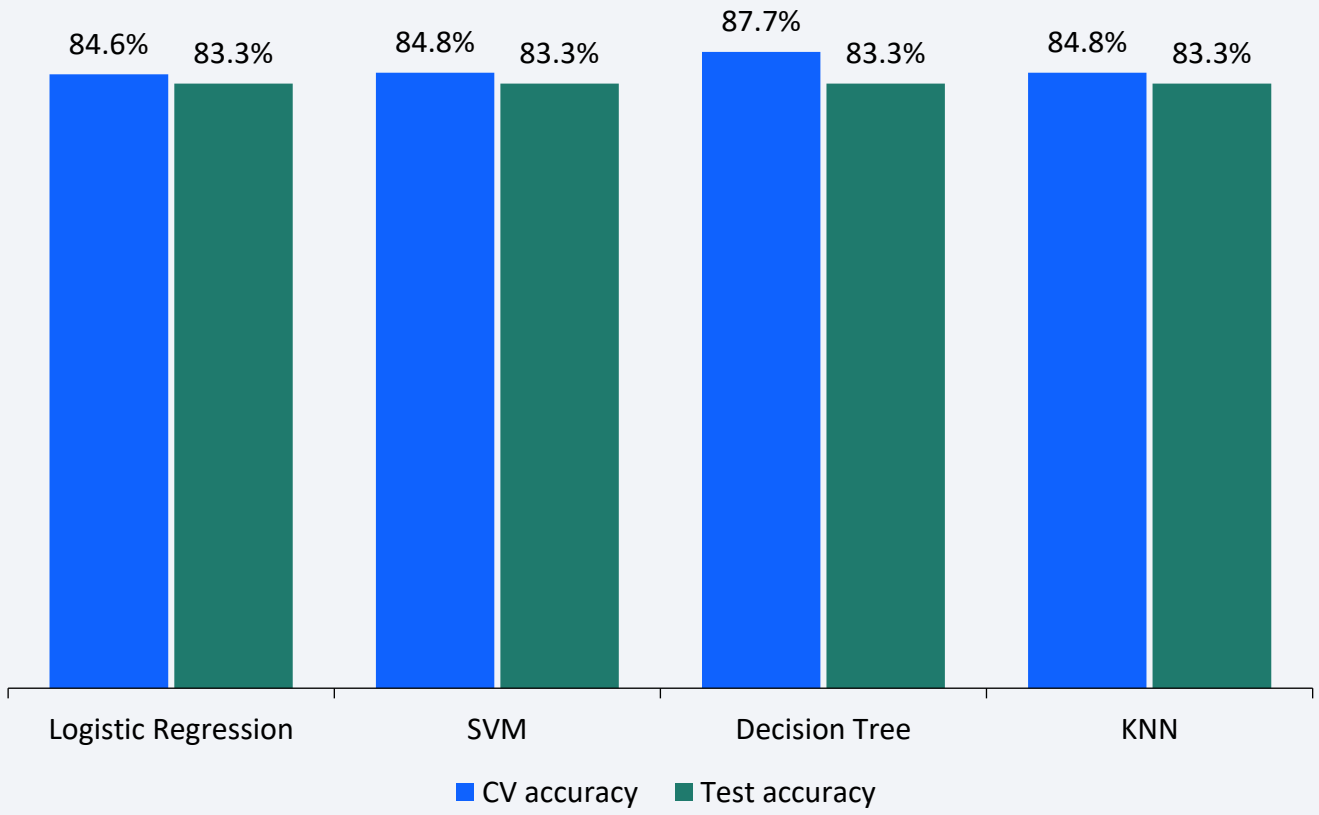
- FT has the best rate among large groups: 66.7% of 24
- v1.0 and v1.1: 1 of 20
- 2,500–5,000 kg: 57.1% of 21
- 5,000–7,500 kg: 22.2% of 9
- Booster version seems to matter more than payload here

Section 5

Predictive Analysis (Classification)

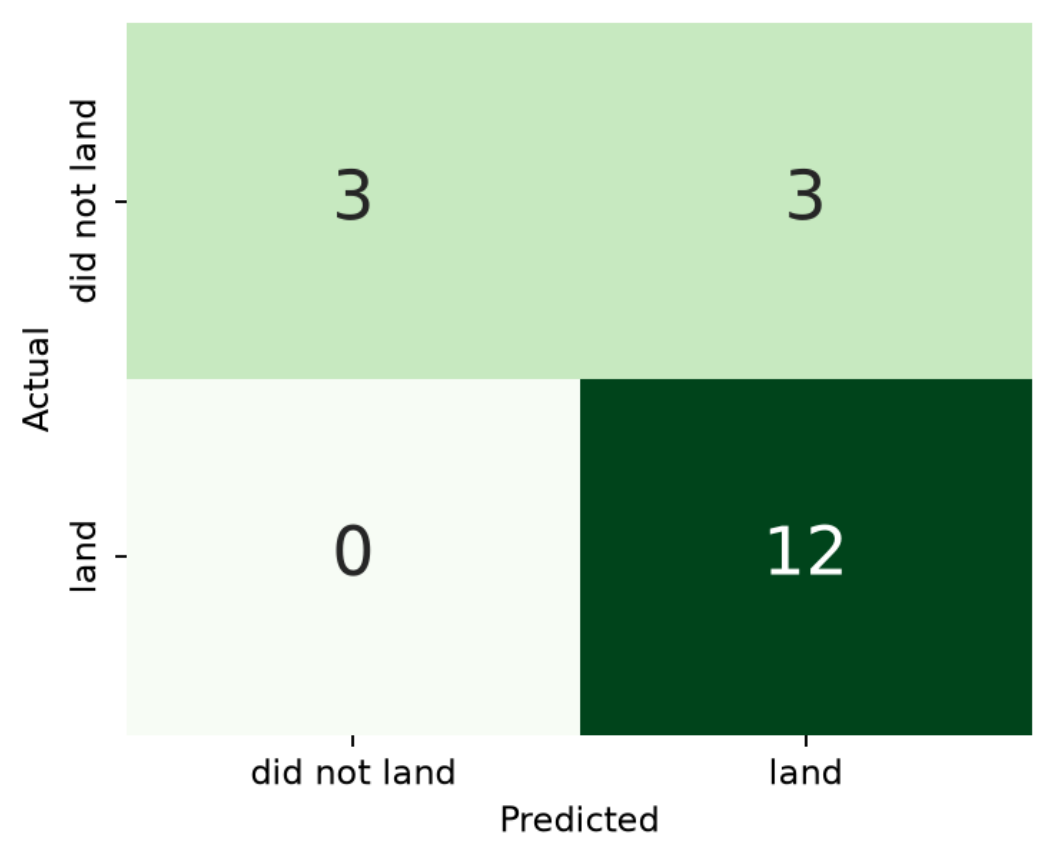


Classification Accuracy



- All four models score 83.3% on the test set: 15 of 18
- Decision Tree has the best CV accuracy: 87.7%
- Best SVM kernel: sigmoid, at 84.8%
- A tie on 18 flights doesn't prove the models are equal

Confusion Matrix



- Decision Tree, the top CV model
- **12 of 12** landings were caught
- **3 of 6** failures were called landings
- Precision for landing: 80%
- False positives are the costly error: we would expect a cheap launch that isn't

Stress Test: 100 Random Splits

Model	Mean	5th–95th pct.	Without LandingPad
KNN	84.7%	72.2–94.4%	84.6%
SVM	84.7%	72.2–94.4%	83.3%
Logistic Regression	83.8%	72.2–94.4%	83.9%
Decision Tree	81.2%	66.7–94.4%	80.4%
Baseline	66.7%	66.7–66.7%	66.7%

Models keep the parameters found by grid search. Each split holds out 18 flights and keeps the class balance.

What this shows

The CV winner, Decision Tree, falls to last.
Accuracy swings about 22 points between unlucky and lucky splits.
Dropping LandingPad barely changes scores, so that field isn't driving results.

Innovative Insights

Small test sets hide differences

With 18 flights, each error is worth 5.6 points. All four models tied, and only repeated splits separated them.

Payload isn't just a proxy for time

Heavy flights beat lighter ones in both eras. After flight 45: 84% of 19 vs. 69% of 26.

Sparse categories can mislead

Four orbits have one flight each, so their 0% and 100% bars are not evidence.

Price the errors differently

A false 'will land' underprices a bid. Tune for precision on landings rather than accuracy.

Conclusions

- Landing success rose from 0% to about 84–90% as the program matured
- KSC LC-39A has the best dashboard record, but sites were used at different stages
- Heavier flights landed more often, while GTO stayed the hardest common orbit
- Models beat the 66.7% baseline, by about 17 points on average
- Logistic Regression is the practical pick: simple, and within 1 point of the best

Next step: test the model on newer launches before using it to price bids.

Appendix

- **Data files:** IBM Skills Network datasets for the SQL, geolocation and dashboard labs
- **Repeat runs:** the Decision Tree uses random_state=2, so results here can differ from a first run in the lab
- **Stress test:** scaling is fitted inside each split, but parameters come from the first grid search, so scores may be slightly optimistic
- **Proximity points:** OpenStreetMap, via Overpass and Nominatim

GitHub: https://github.com/MateoGomezTamayo/Data-Science-knowledge-base/tree/main/10_portfolio/21_spacex_capstone

Thank you!

